Jorge's Little Game

Tutorial for editing the JSON file(s):

General info:

We have two JSON files, one called “gameParameters.json”, found inside the build Folder address/GraFunPlatformer_Data/StreamingAssets.
The second JSON file is inside a folder called “Extras” , and it’s called “extrasGameParameters.JSON”

The first one WILL be read and used by the game, the second one is optional, along with the pngs included along is within the “Extras” folder.

The JSON files let the game know how to use/interpret the different sprite assets (png format), for best performance and to best use your art. They also allow for some extra gameplay tweaking.

To use the JSON files:

Be careful not to modify the structure of the JSON document, if you cut, replace of misplace a {, }, [or], the system won’t be able to read it and you’ll get mistakes.

Every { is paired with a }, every [has a].

Every piece of data the game uses has a field name/identifier between quotation marks “ “, before a colon : and the data, and each piece is separated with comma’s

Example: “width”: 1.64,

Some pieces of data are numbers, some is text. Whenever the data is text, it will also be between quotation marks “ “. Numbers won’t.

Example: “character” : “Screen”,

“yOffset”: -0.13,

Do not change the field names/identifiers... if the game doesn’t recognize them, you’ll get errors.
You can’t add new ones. The game won’t recognize them and you’ll get errors.

If a file fails to load, the game will show an error screen naming what’s wrong - it won’t silently fail/freeze. Use the error screen’s info to detect any errors you may have.

For clearer organization, the JSON data fields are group within containers with descriptive titles.

Example: “gameplayTuning”{ ←This is just the container, and everything between this curly bracket will be part of it

JSON files content:

Following you'll get the structure of the "gameParameters.JSON" file, so you know how to tune the json

Whenever a type is listed as number, you can use decimal points. If it's listed as "Integer", you have to use whole numbers without decimal point

gameParameters.json content

Field name/identifier	Data type	What it does	Notes
debugging		NO DATA: Container for organization, everything related to debugging	
debugging/drawDebugGizmos	boolean (true/false)	Draws the real hitboxes (player, enemy sight/body/head) as outlines in-game, so you know how the numbers in the following section are actually producing.	
gameplayTuning		NO DATA: Container for organization, everything related to gameplay tuning.	
gameplayTuning/playerCollisionBox	{width, height, yOffset}	The player's physical hitbox. yOffset nudges it vertically; positive is up. (**yOffset will be consisent across every field including a yOffset**)	
gameplayTuning/enemySightBox	{width, height, yOffset}	How far an enemy can "see" the player. Only matters for an enemy that has an attack (see Extras) : an enemy with no attack art ignores this and just patrols.	
gameplayTuning/enemyBodyBox	{width, height, yOffset}	The enemy's damaging hitbox: touching it hurts the player.	
gameplayTuning/enemyHeadBox	{height}	The enemy's stompable hitbox, sitting on top of enemyBodyBox. Only height is adjustable: width always matches the body box.	
gameplayTuning/stompDamage	number	Damage dealt to an enemy when the player stomps it.	
gameplayTuning/enemyHealth	number	An enemy's max HP.	
gameplayTuning/cameraDistance	number (range: 5–7)	Camera zoom. 7 = Most zoomed out, 5 = Most zoomed in. Anything outside 5–7 is automatically clamped back into range: you'll never get an error for this one.	
blendModes		NO DATA: Container for organization, everything related to blendModes	

blendModes/character	“Normal" / "Lighten" / "Screen" / "Multiply	Blend mode for the player sprite.	An unrecognized or missp falls back to Normal: checl mode doesn't seem to app
blendModes/enemy	“Normal" / "Lighten" / "Screen" / "Multiply	Blend mode for enemy sprites.	An unrecognized or missp falls back to Normal: checl mode doesn't seem to app
anims & animsCollectible		NO DATA: Container for organization, everything related to animations	Each animation within thes be linked to the pngs with the StreamingAssets folde names in the JSON don’t c game won’t recognize ther corresponding pngs have matching title (case sensit ALL animations in anim & should have a correspondi same title. The required an anims: characterRun, char characterIdle, enemyIdle animsCollectible: objColle
name	string	Must exactly match one of the required names above: this is how the game finds the entry, not the array order. **(You shouldn’t change this one)**	
yOffset	number	Vertical nudge for this sprite, in world units.	
columns / rows	integer	Grid size of your sprite sheet image: How many frames does your png have vertically/horizontally.	Please note that the grid s number of grid spaces you and not the actual number inside it.
numFrames	integer	How many cells of that grid are real frames: see Sprite sheet rules.	
frameSpeed	number	Seconds each frame is shown. The smaller the number, the faster the animation will play.	
scaleX / scaleY	number, default 1	Render scale multiplier.	Your sprites should be mad scale inside photoshop, bu the scale here if needed.
loopUpStartsAt / loopUpEndsAt	integer, default -1 (off)	characterJump only: frame range to hold while rising.	You don’t have to include a loop will allow for smoothe
loopDownStartsAt / loopDownEndsAt	integer, default -1 (off)	characterJump only: frame range to hold while falling.	You don’t have to include a loop will allow for smoothe

extrasGameParameters.json content

Field name/identifier	Data type	What it does	Notes

anims			Each animation in the file h than the regular mandatory “projectileReleaseFrame”. F fields, please refer to the pr “gameParameters.json con
projectileReleaseFrame	integer, default 0 (off)	The frame number that fires a projectile. Needs a matching entry in animsProjectile below to actually have something to fire.	
animsProjectile	characterProjectile/enemyProjectile		Same core fields as anims (rows, numFrames, frameSp extra fields of its own. For i please refer to the previous “gameParameters.json con

Information about how to create the sprite sheets (relevant also for the json files, and how to fill the animation data):

Frames are read left to right, then top to bottom — like reading a page. A 2×2 sheet with 4 frames reads: top-left, top-right, bottom-left, bottom-right.

numFrames can be less than columns × rows if you have empty cells at the end of the sheet — just leave them blank and set numFrames to however many real frames you actually drew.

Tip: You can also have empty frames in between your animation frames to create a flicker effect, if that makes sense for your animation.

CAREFUL: Soft, anti-aliased edges can **glow** under Lighten/Screen/Multiply blend modes. If your sprite's background was removed from a white (or colored) canvas, a faint rim of that color can survive at the edge. It's invisible in Normal mode, but Lighten/Screen/Multiply don't fade it the same way, so keep edges clean (or fully transparent, not just low-opacity) if you plan to use one of those blend modes.

Which json-related files to include:

Required (Note: this list only includes the pngs that are referenced/linked to the json, all environment assets are mandatory, even if not listed here):

characterRun.png
characterJump.png
characterIdle.png
enemyIdle.png
objCollectible.png

Optional:

Extras/characterAttack.png
Extras/enemyAttack.png
Extras/characterProjectile.png
Extras/enemyProjectile.png
Extras/healthBar.png

Troubleshooting:

Game shows an error screen on launch: a JSON file has a syntax mistake (missing comma, unmatched bracket, missing quote). The screen will show which file and what's wrong.

An animation looks scrambled: your columns/rows/numFrames don't match the actual sheet. Recount the grid in your image.

A blend mode doesn't seem to apply: check spelling and capitalization; it must be exactly Normal, Lighten, Screen, or Multiply (if it’s misspelled it will default to Normal, and silently fail to use the one you wanted).

Camera zoom doesn't seem to go far enough: that's expected; cameraDistance is intentionally clamped to 5–7.

An attack/projectile/health bar never appears: that's just the corresponding optional file missing (or the name inside the JSON not matching exactly).